EXPERIMENT 6

DATA VISUALIZATION USING POWER BI

Aim

To import, transform, analyze, and visualize data using Power BI Desktop and create interactive dashboards and reports for decision-making.

Software Requirements

• Microsoft Power BI Desktop

• Microsoft Excel

• CSV Dataset

• Windows Operating System

Algorithm 1

Importing Data into Power BI

Step 1

Open Power BI Desktop.

Step 2

Select Home → Get Data.

Step 3

Choose Text/CSV.

Step 4

Browse and select the dataset file 231401057.csv.

Step 5

Click Load.

Step 6

Verify that the artist streaming dataset is imported correctly in Power BI.

Dataset used: Artist Streaming Dataset (500 records, 14 fields).

Important fields include Artist Name, Country of Origin, Primary Genre, Artist Type, Debut Year, Total Streams (in millions), Lead Streams (in millions), Feature Streams (in millions), Solo Streams (in millions), and Collaborative Streams (in millions).

Algorithm 1 – Output

Imported Dataset

The artist streaming dataset was successfully imported into Power BI.

Artist Name	Country of Origin	Primary Genre	Total Streams (in millions)
Drake	Canada	Hip-Hop	137,492.1
Taylor Swift	United States	Pop	127,861.0
Bad Bunny	United States	Reggaeton	125,899.8

The imported data contains artist, demographic, genre, debut-year, and streaming-related attributes.

Algorithm 2

Data Transformation using Power Query

Step 1

Select Transform Data.

Step 2

Open Power Query Editor.

Step 3

Check the dataset and remove duplicate records if present.

Step 4

Check for null or missing values and handle them appropriately.

Step 5

Verify and rename columns where necessary; retain the dataset field names used for visualization.

Step 6

Select Close & Apply to load the transformed data into Power BI.

Output

Cleaned artist streaming dataset loaded successfully into Power BI.

Algorithm 3

Creating a Bar Chart

Step 1

In Report View, select the Clustered Column Chart visualization.

Step 2

Drag Primary Genre → X-axis.

Step 3

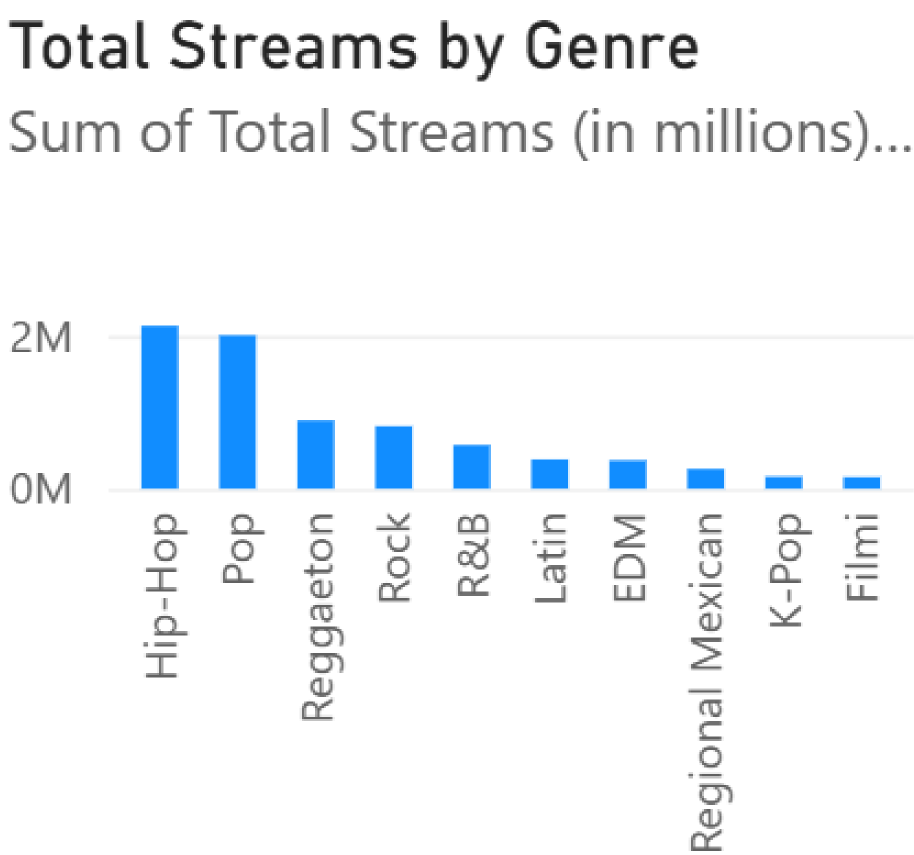
Drag Total Streams (in millions) → Y-axis.

Step 4

Open Format visual and set the chart title to Total Streams by Genre.

Output Visualization

Total Streams by Genre



Interpretation

Hip-Hop has the highest total streams among the genres in the dataset (2,126,055.1), while Nu Metal has the lowest total streams (7,018.9).

Algorithm 4

Creating a Pie Chart

Step 1

Select the Pie Chart visualization.

Step 2

Drag Primary Genre → Legend.

Step 3

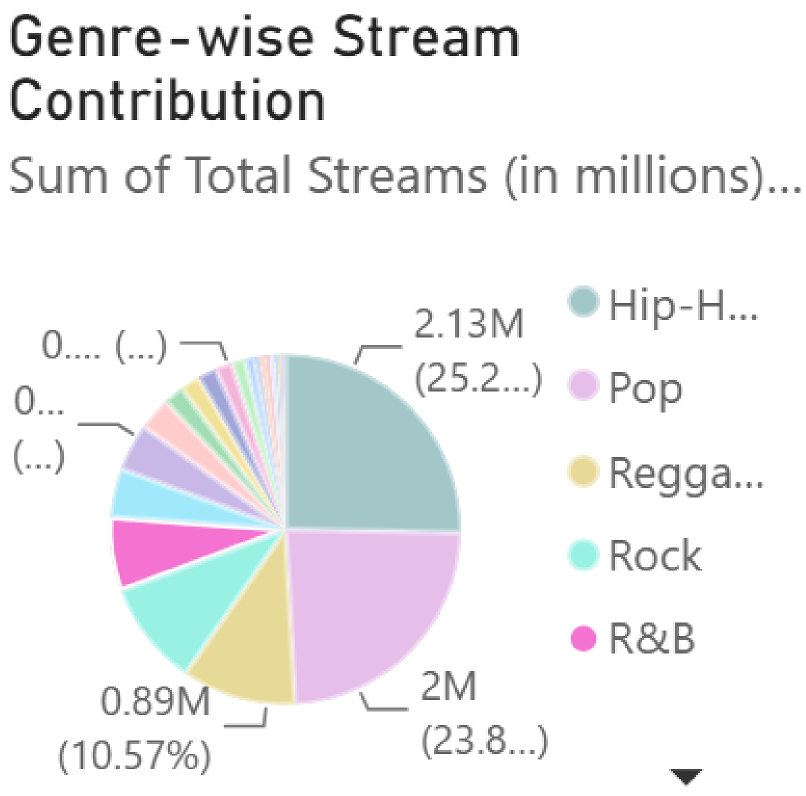
Drag Total Streams (in millions) → Values.

Step 4

Enable Detail labels/Data labels and set the chart title to Genre-wise Stream Contribution.

Output Visualization

Genre-wise Stream Contribution



Interpretation

The pie chart shows the percentage contribution of each genre to total streams. Hip-Hop contributes the largest share, approximately 25.28% of the total.

Algorithm 5

Creating a KPI Card

Step 1

Select the Card visualization.

Step 2

Drag Total Streams (in millions) → Values.

Step 3

Apply formatting and set the title to Total Streams KPI.

DAX Formula

Total Streams

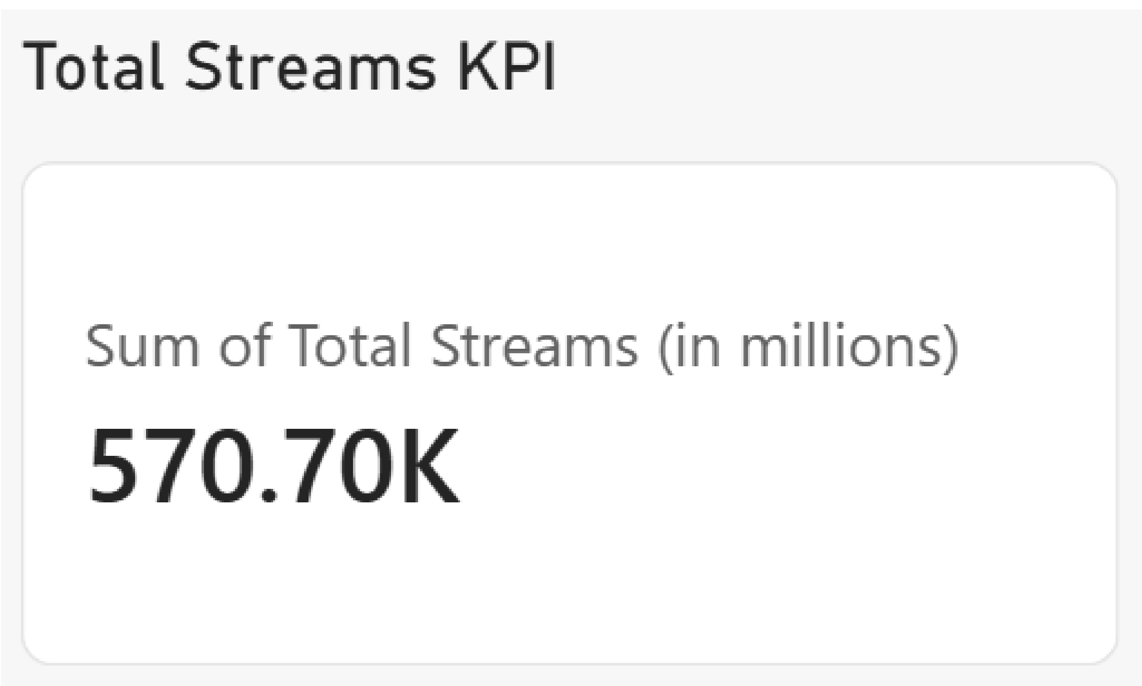
Total Streams = SUM('231401057'[Total Streams (in millions)])

DAX Measure

Total Streams = SUM('231401057'[Total Streams (in millions)])

Output

KPI card displaying the sum of Total Streams. The uploaded screenshot shows 570.70K under its current filter context.



Algorithm 6

Creating Slicers

Step 1

Select the Slicer visualization.

Step 2

Drag Country of Origin → Field.

Step 3

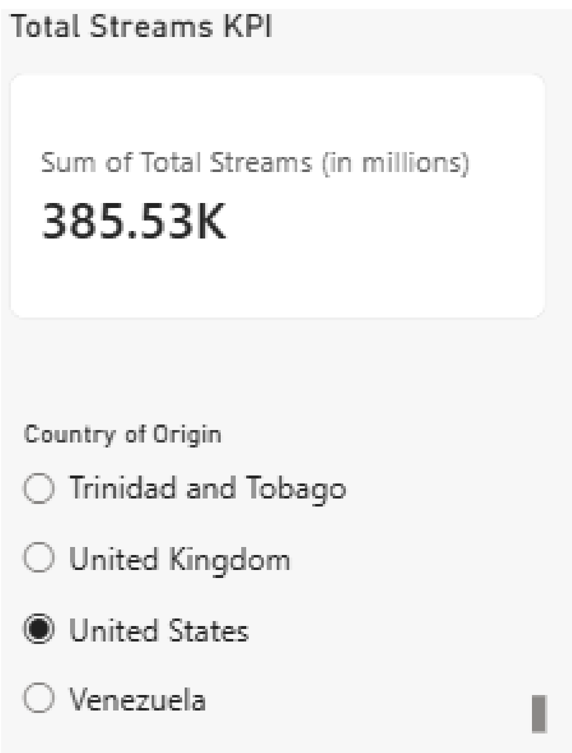
Enable Single select if only one country should be selected at a time.

Step 4

Select a country and verify that the KPI and other visuals update according to the selected filter.

Output

Country of Origin slicer used to filter the Power BI visuals. The screenshot shows United States selected, with the KPI updating to 385.53K.



Algorithm 7

Creating DAX Calculations

Average Streams

Average Streams = AVERAGE('231401057'[Total Streams (in millions)])

Maximum Streams

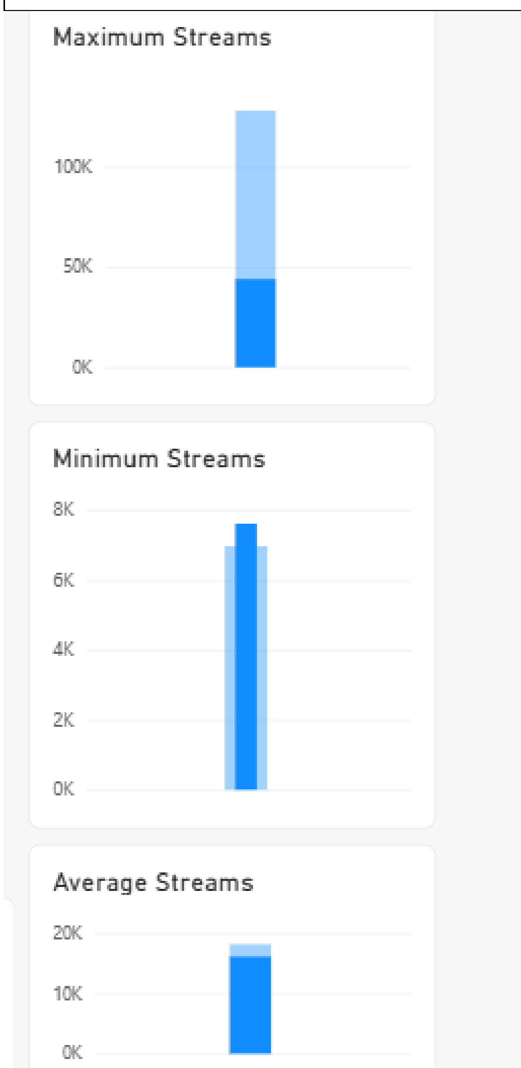
Maximum Streams = MAX('231401057'[Total Streams (in millions)])

Minimum Streams

Minimum Streams = MIN('231401057'[Total Streams (in millions)])

Output

Measure	Value
Average Streams	16,817.61
Maximum Streams	137,492.10
Minimum Streams	6,972.00



Algorithm 8

Creating Dashboard

Step 1

Use the previously created KPI Card.

Step 2

Use the previously created Bar Chart.

Step 3

Use the previously created Pie Chart.

Step 4

Use the previously created Country of Origin Slicer.

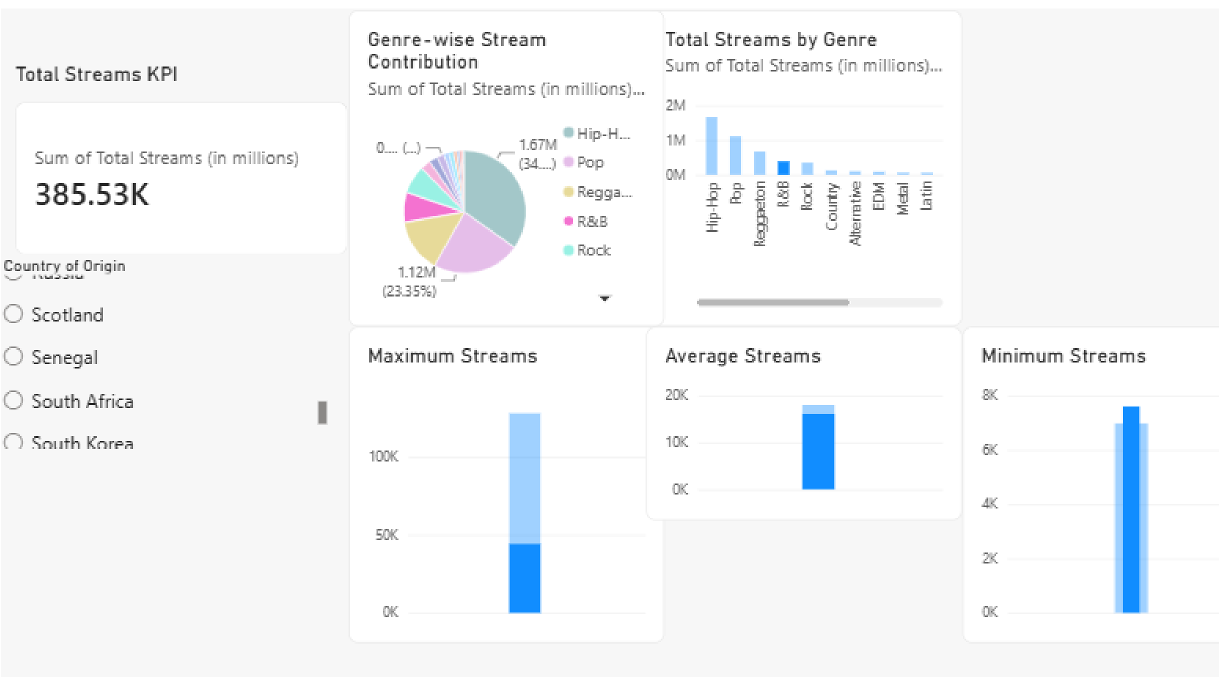
Step 5

Arrange the existing visuals neatly on the report canvas without overlapping.

Step 6

Test the slicer to confirm that the visuals respond to the selected country, then save the Power BI file.

Dashboard Layout



Result:

Data visualization using Power BI has been done successfully using the artist streaming dataset.